

Introduction to Scaling Laws

Instructor: Hyunwoo J. Kim

Today's agenda

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1. Motivation
2. Definition of Scaling Laws
3. History Before Scaling Laws
4. Power-Law Scaling & Empirical Results
5. Limitation of Power-Law Scaling

1. Motivation

Rocket Science Analogy

- **Rocket Science:**
 - One-shot, high-cost experiments with limited opportunity for iteration
 - Small-scale rockets and subsystems are tested first, and physical laws guide extrapolation to full-scale designs
- **Old (naïve): Tune hyperparameters directly on large models**
 - Inefficient in time and computational cost
- **New (optimistic?): Tune on small models**
 - extrapolate to large-scale models
 - Using **Scaling Laws**



Scaling in Practice: Large Language Models

- Scaling laws has become a **standard analytical lens** in modern LLM research

GPT-3

Chinchilla

DeepSeek

Language Models are Few-Shot Learners

Tom B. Brown*	Benjamin Mann*	Nick Ryder*	Melanie Subbiah*	
Jared Kaplan ¹	Prafulla Dhariwal	Arvind Neelakantan	Pranav Shyam	Girish Sastry
Amanda Askell	Sandhini Agarwal	Ariel Herbert-Voss	Gretchen Krueger	Tom Henighan
Rewon Child	Aditya Ramesh	Daniel M. Ziegler	Jeffrey Wu	Clemens Winter
Christopher Hesse	Mark Chen	Eric Sigler	Mateusz Litwin	Scott Gray
Benjamin Chess		Jack Clark	Christopher Berner	
Sam McCandlish	Alec Radford	Ilya Sutskever	Dario Amodei	

OpenAI



Training Compute-Optimal Large Language Models

Jordan Hoffmann*, Sébastien Borgeaud*, Arthur Mensch*, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals and Laurent Sifre*

*Equal contributions

We investigate the optimal model size and number of tokens for training a transformer language model under a given compute budget. We find that current large language models are significantly under-trained, a consequence of the recent focus on scaling language models whilst keeping the amount of training data constant. By training over 400 language models ranging from 70 million to over 16 billion parameters on 5 to 500 billion tokens, we find that for compute-optimal training, the model size and the number of training tokens should be scaled equally: for every doubling of model size the number of training tokens should also be doubled. We test this hypothesis by training a predicted compute-optimal model, Chinchilla, that uses the same compute budget as Gopher but with 70B parameters and 4x more data. Chinchilla uniformly and significantly outperforms Gopher (280B), GPT-3 (175B), Jurassic-1 (178B), and Megatron-Turing NLG (530B) on a large range of downstream evaluation tasks. This also means that Chinchilla uses substantially less compute for fine-tuning and inference, greatly facilitating downstream usage. As a highlight, Chinchilla reaches a state-of-the-art average accuracy of 67.5% on the MMLU benchmark, greater than a 7% improvement over Gopher.



DeepSeek LLM

Scaling Open-Source Language Models with Longtermism

Xiao Bi, Deli Chen, Guanting Chen, Shanhuang Chen, Damai Dai, Chengqi Deng, Honghui Ding, Kai Dong, Qiushi Du, Zhe Fu, Huazuo Gao, Kaige Gao, Wenjun Gao, Ruiqi Ge, Kang Guan, Daya Guo, Jianzhong Guo, Guangbo Hao, Zhewen Hao, Ying He, Wenjie Hu, Panpan Huang, Erhang Li, Guowei Li, Jiashi Li, Yao Li, Y.K. Li, Wenzeng Liang, Fangyun Lin, A.X. Liu, Bo Liu, Wen Liu, Xiaodong Liu, Xin Liu, Yiyuan Liu, Haoyu Lu, Shanghao Lu, Fuli Luo, Shirong Ma, Xiaotao Nie, Tian Pei, Yishi Piao, Junjie Qiu, Hui Qu, Tongzheng Ren, Zebui Ren, Chong Ruan, Zhangli Sha, Zhihong Shao, Junxiao Song, Xuecheng Su, Jingxiang Sun, Yaofeng Sun, Minghui Tang, Bingxuan Wang, Peiyi Wang, Shiyu Wang, Yaobai Wang, Yongji Wang, Tong Wu, Y. Wu, Xin Xie, Zhenda Xie, Ziwei Xie, Yiliang Xiong, Hanwei Xu, R.X. Xu, Yanhong Xu, Dejian Yang, Yuxiang You, Shuiping Yu, Xingkai Yu, B. Zhang, Haowei Zhang, Lecong Zhang, Liyue Zhang, Mingchuan Zhang, Minghua Zhang, Wentao Zhang, Yichao Zhang, Chenggang Zhao, Yao Zhao, Shangyan Zhou, Shunfeng Zhou, Qihao Zhu, Yuheng Zou*

*DeepSeek-AI

- [1] Brown, Tom B., et al. "Language Models are Few-Shot Learners." NeurIPS, 2020. [2] Hoffmann, Jordan, et al. "Training Compute-Optimal Large Language Models." NeurIPS, 2022. [3] DeepSeek-AI. "DeepSeek LLM: Scaling Open-Source Language Models with Longtermism." *arXiv*, 2024.

Imagine...

You are given **ten thousand (1024) H100s** for **6 months** and you need to build a good open-source LM.

Scaling Laws in a Resource-Constrained AI Race

- WBL (World Best LLM) involves national-scale AI development under **fixed timelines and limited GPU resources**
- In this setting, large-scale training is expensive and hard to repeat, making design mistakes **costly and irreversible**
- Scaling laws** enable performance prediction before full-scale training

국가대표 AI 모델 선정된 기업들 (자료: 과학기술정보통신부)

네이버클라우드

참여 기관 네이버, 트윅브랩스, 서울대 산학협력단, 카이스트, 포항공대 산학협력단, 고려대 산학협력단, 한양대 산학협력단
주요 과제 범국민 접근성 확대와 산업 확산을 위한 독자 오픈파운데이션 모델 구축

SK텔레콤

참여 기관 크래프톤, 포티투닷, 리벨리온, 라이너, 셀렉트스타, 서울대 산학협력단, 카이스트
주요 과제 포스트-트랜스포머 AI 모델로 K-AI 서비스 구현

LG AI연구원

참여 기관 LG윙플러스, LG CNS, 이스트소프트, 프렌들리AI, 이스트에이드, 한글과컴퓨터, 슈퍼브AI, 퓨리오사AI, 휘튼테크놀로지스
주요 과제 글로벌 최고 수준의 프론티어 AI 모델 개발

업스테이지

참여 기관 노타, 래블업, 플리토, 뷰노, 마카나락스, 로엔컴퍼니, 오케스트로, 데이원컴퍼니, 울가나이즈코리아, 금융결제원, 서강대 산학협력단, 카이스트
주요 과제 글로벌 프론티어 수준의 독자 AI 파운데이션 모델 '솔라 WBL' 개발

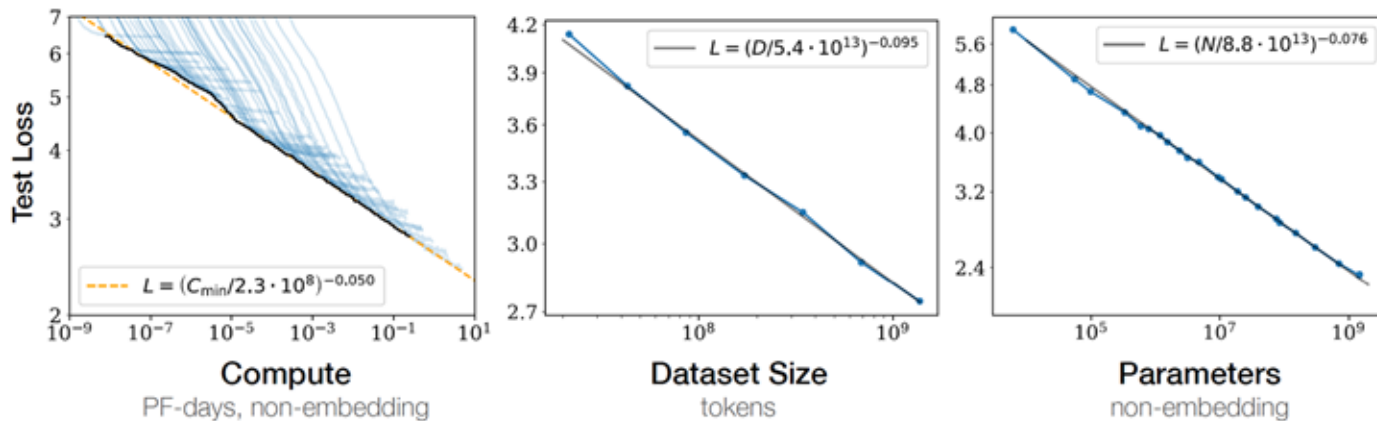
NCAI

참여 기관 고려대, 서울대, 연세대, 카이스트, 한국전자통신연구원, 에이아이웍스, 포스코DX, 롯데이노베이션, HIL로보틱스, 인타엑스, 미디어젠, 문화방송, NHN
주요 과제 독자 대규모 언어 파운데이션 모델 패키지 개발, 도메인집중 플랫폴 구축

2. Definition of Scaling Laws

What is Scaling Law?

- **A Scaling Law** describes how model loss or error changes as the amount of training resources increases
- **Scale (resources)**: Compute, dataset size, and model parameters
- **What it captures**: A consistent performance trend across scales, typically following a **power-law relationship**



Power-Law Form and Log-Log Interpretation

- Empirical form

$$L = L_0 + \left(\frac{x_0}{x} \right)^\alpha$$

x : training resource (compute, dataset size, or model parameters)

x_0 : reference scale of the resource

α : scaling exponent

L_0 : irreducible loss

Power-Law Form and Log-Log Interpretation

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Log-Log Interpretation

$$L = L_0 + \left(\frac{x_0}{x} \right)^\alpha$$



Log-Log Transform!

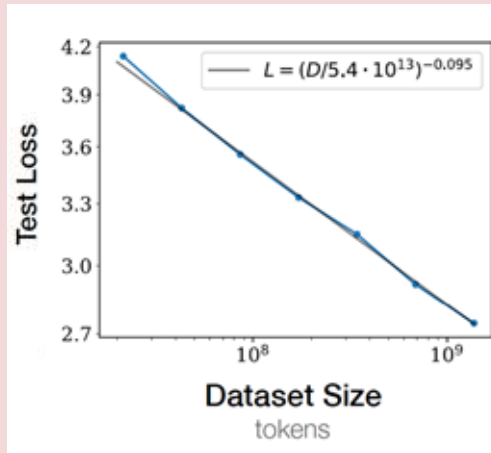
$$\log(L - L_0) = \alpha \log x_0 - \alpha \log x$$

x : training resource (compute, dataset size, or model parameters)

x_0 : reference scale of the resource

α : scaling exponent

L_0 : irreducible loss



Ex) Loss and dataset size is linear on a log-log plot

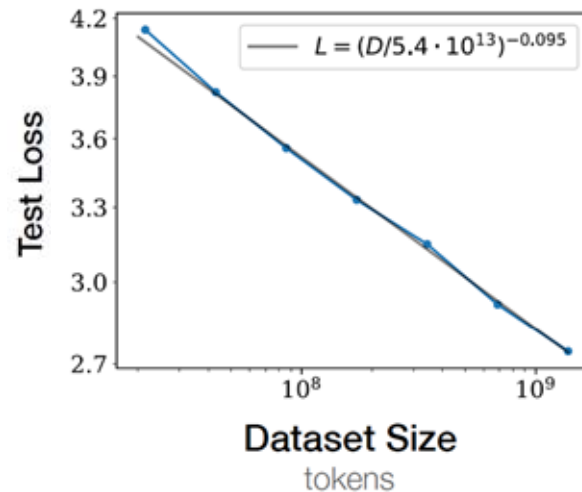
- In log-log space, the power-law form becomes **linear**
- Straight lines in log-log space indicate predictable improvement with scale
- The slope of the line corresponds to the scaling exponent

Single-Resource Scaling vs Joint Scaling

Aspect	Single-Resource Scaling	Joint Scaling
Analysis focus	Isolating the effect of a single resource	Capturing interactions between resources
Empirical form	$L = L_0 + \left(\frac{x_0}{x} \right)^\alpha$	From Rosenfeld+ 2020 $L(n, m) = A \cdot n^{-\alpha} + B \cdot m^{-\beta} + L_0$ m (dataset size), n (model size)
Log-log behavior	Straight line	Planar linear structure

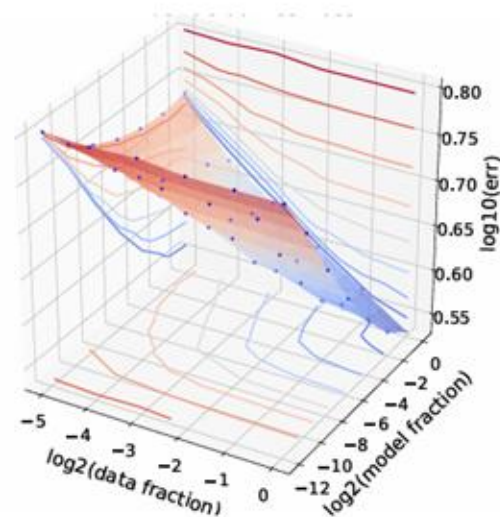
Single-Resource Scaling

Aspect	Single-Resource Scaling
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Joint Scaling

Aspect	Joint Scaling
Analysis focus	Capturing interactions between resources
Empirical form	From Rosenfeld+ 2020 $L(n, m) = A \cdot n^{-\alpha} + B \cdot m^{-\beta} + L_0$ m (dataset size), n (model size)
Log-log behavior	Planar linear structure



(a) Wiki103 error (cross entropy) landscape.

3. History before Scaling Laws

Theoretical analysis for worst-case bounds

Theorists have thought about ‘scaling’ for a long time...

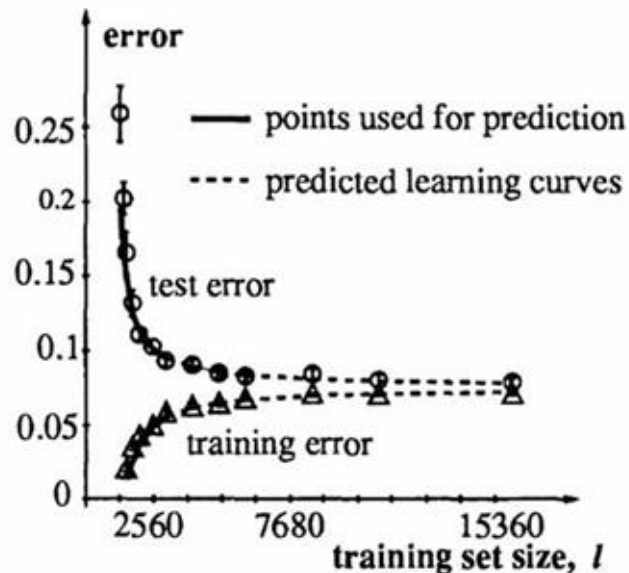
$$\epsilon(\hat{h}) \leq \min_{h \in \mathcal{H}} \epsilon(h) + 2\sqrt{\frac{1}{m} \log \frac{2k}{\delta}}$$

- Early learning theory formalized **how error scales with sample size**
- But focused on **worst-case bounds**, NOT predictive scaling laws

Early empirical evidence of data scaling (pre-formalization)

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[From Cortes & Vapnik (1995)]



$$\varepsilon_{\text{test}} = a + \frac{b}{\ell^\alpha}, \quad \varepsilon_{\text{train}} = a - \frac{c}{\ell^\beta}, \quad \ell: \text{training set size}$$

[From Banko & Brill (2001)]

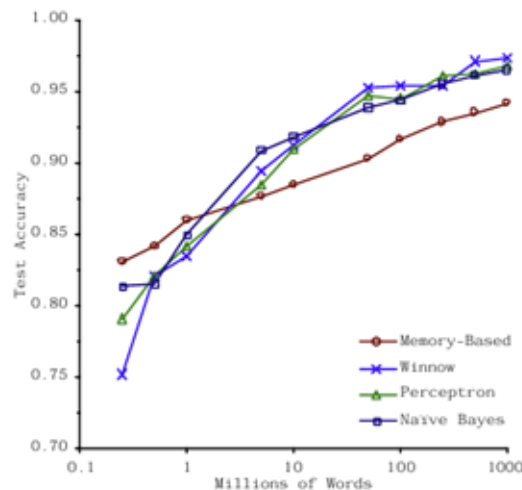


Figure 1. Learning Curves for Confusion Set Disambiguation

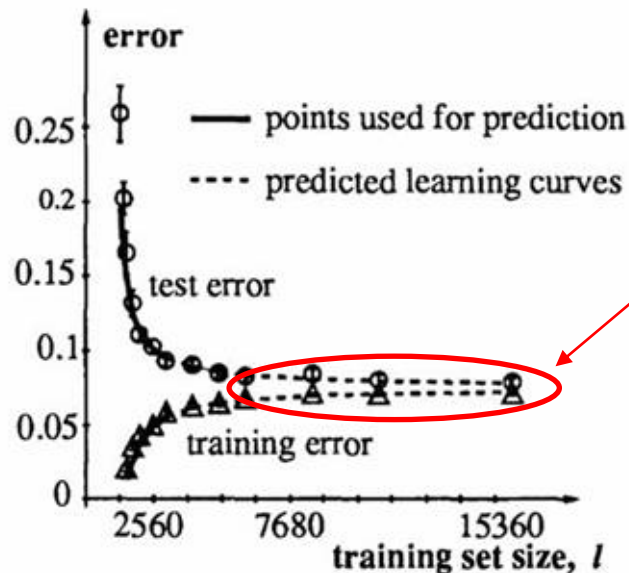
[1] Cortes, Corinna, and Vladimir Vapnik. "Support-Vector Networks." *Machine Learning*, 1995.

[2] Banko, Michele, and Eric Brill. "Scaling to Very Very Large Corpora for Natural Language Disambiguation." *ACL*, 2001.

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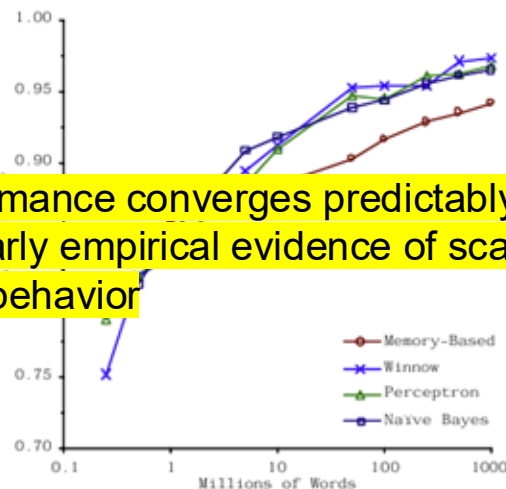


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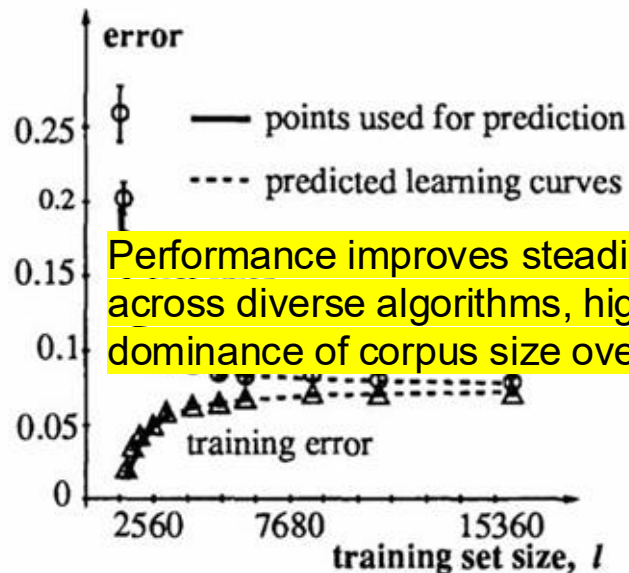
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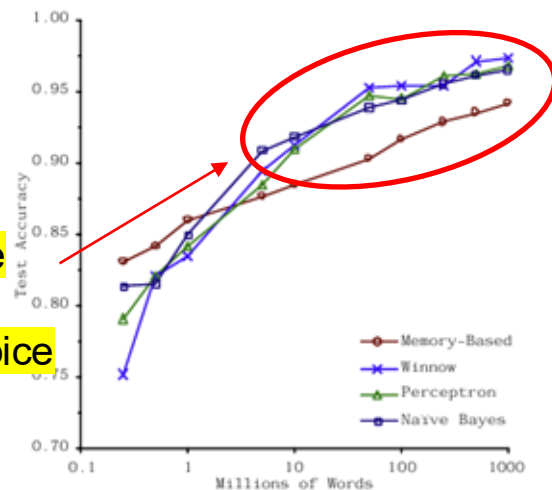


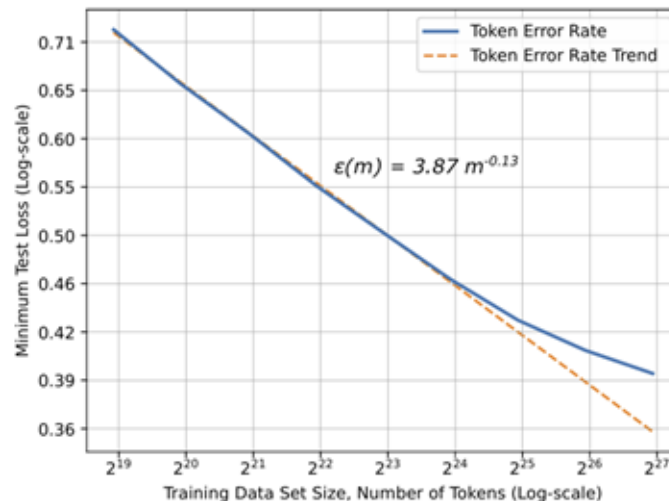
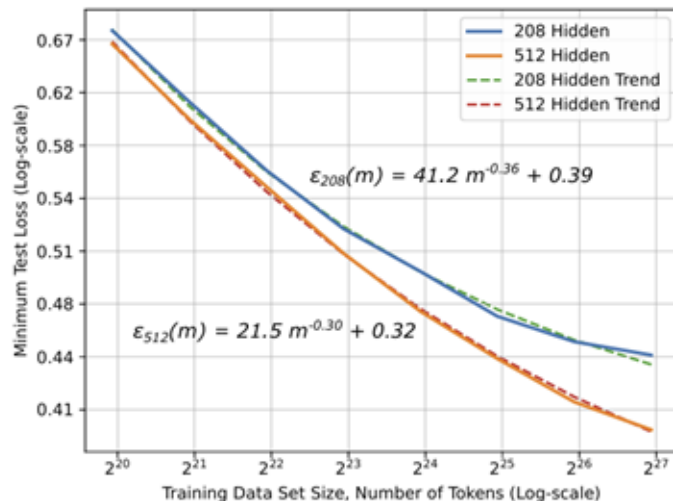
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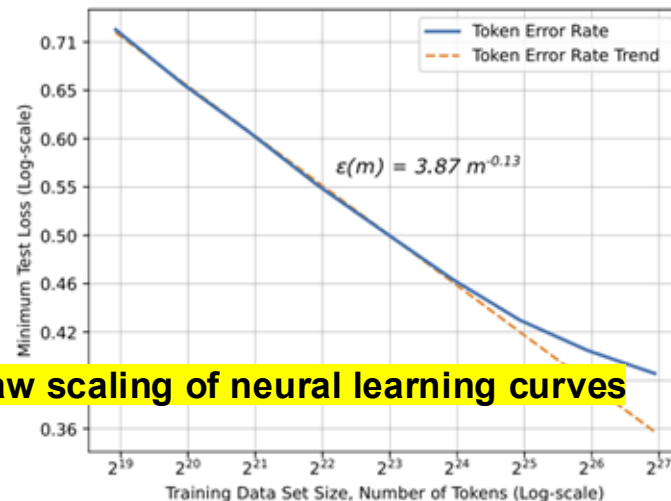
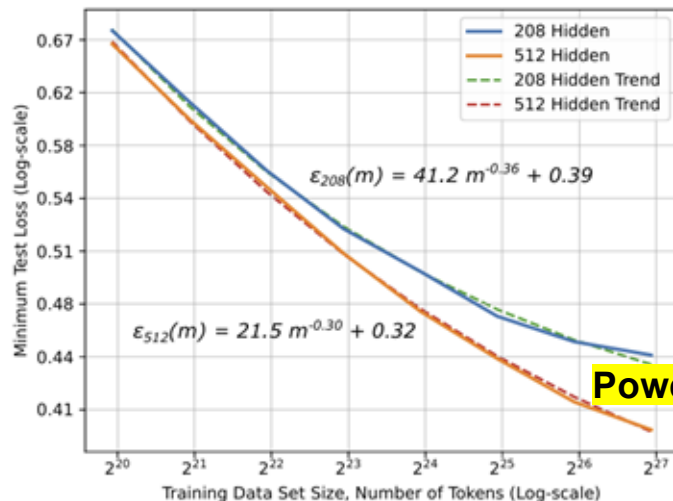
Predictable Scaling in Neural Networks

[From Hestness et al. (2017)]



Predictable Scaling in Neural Networks

[From Hestness et al. (2017)]



Power-law scaling of neural learning curves

4. Why do we observe a power law?

A Toy Example: Power-Law Scaling of MSE

Toy example: mean estimation

Input: $x_1, \dots, x_n \sim \mathcal{N}(\mu, \sigma^2)$

Task: Estimate the average as $\hat{\mu} = \frac{1}{n} \sum_i x_i$

Mean Squared Error: $\mathbf{E}[(\hat{\mu} - \mu)^2] = \frac{\sigma^2}{n}$

A Toy Example: Power-Law Scaling of MSE

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Log-Log Transform!

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Log-Log Transform!

Scaling Behavior: $\log(\text{Error}) = -\log n + 2 \log \sigma$

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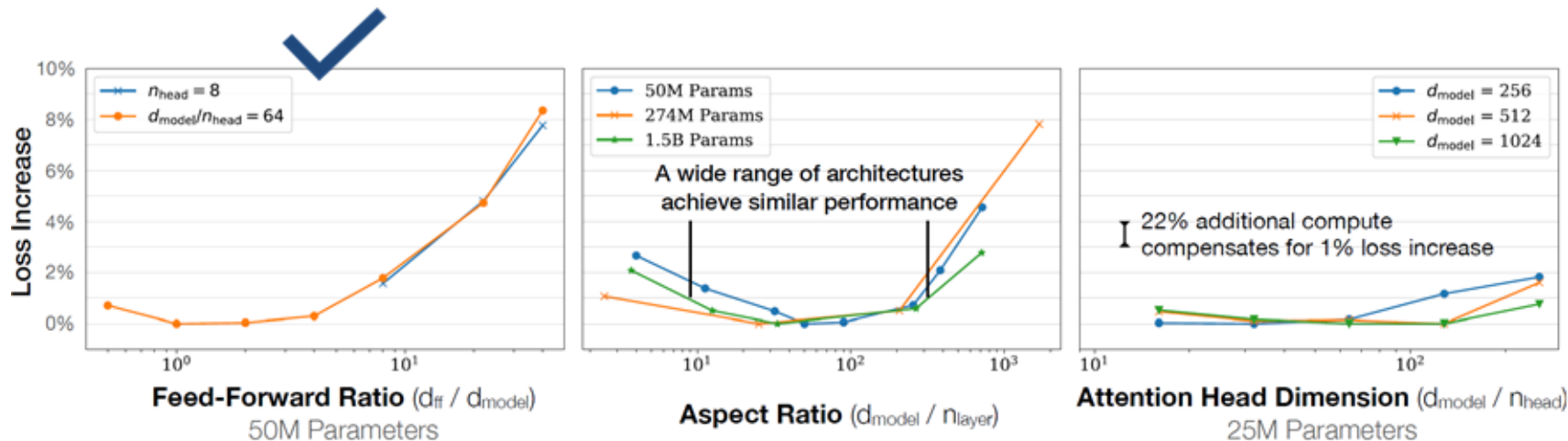
Log-Log Transform!

Scaling Behavior: $\log(\text{Error}) = -\log n + 2 \log \sigma$

Error decreases as a power law in sample size ($\propto 1/n$)

Performance is weakly sensitive to architecture ratios

Feed-Forward Ratio

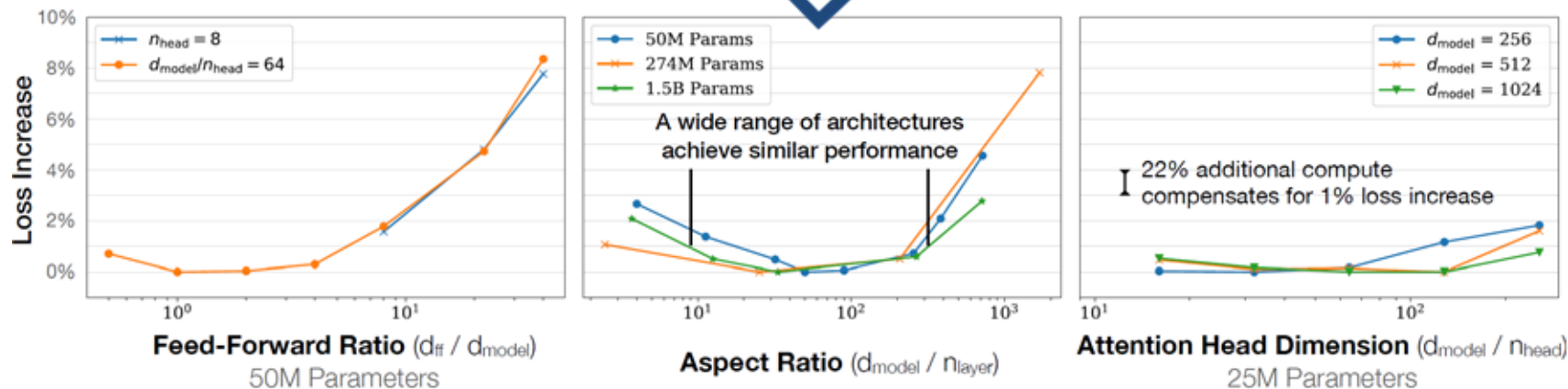


[Kaplan+ 2020]

- Performance is weakly sensitive to the **Feed-Forward ratio** across a wide range of values

Performance is weakly sensitive to architecture ratios

Aspect Ratio

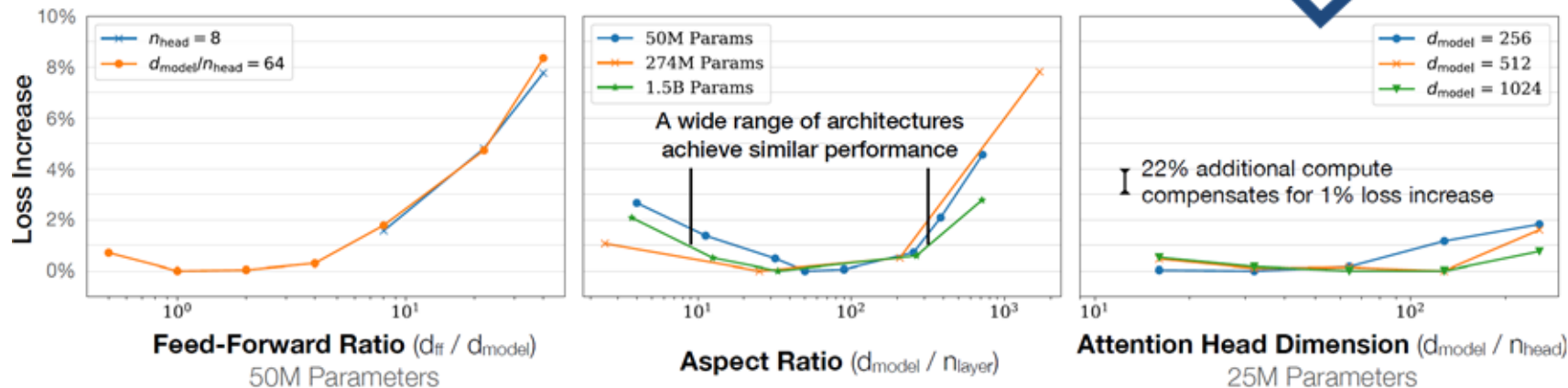


[Kaplan+ 2020]

- Models with different **depth-width aspect ratios** achieve similar performance at the same scale

Performance is weakly sensitive to architecture ratios

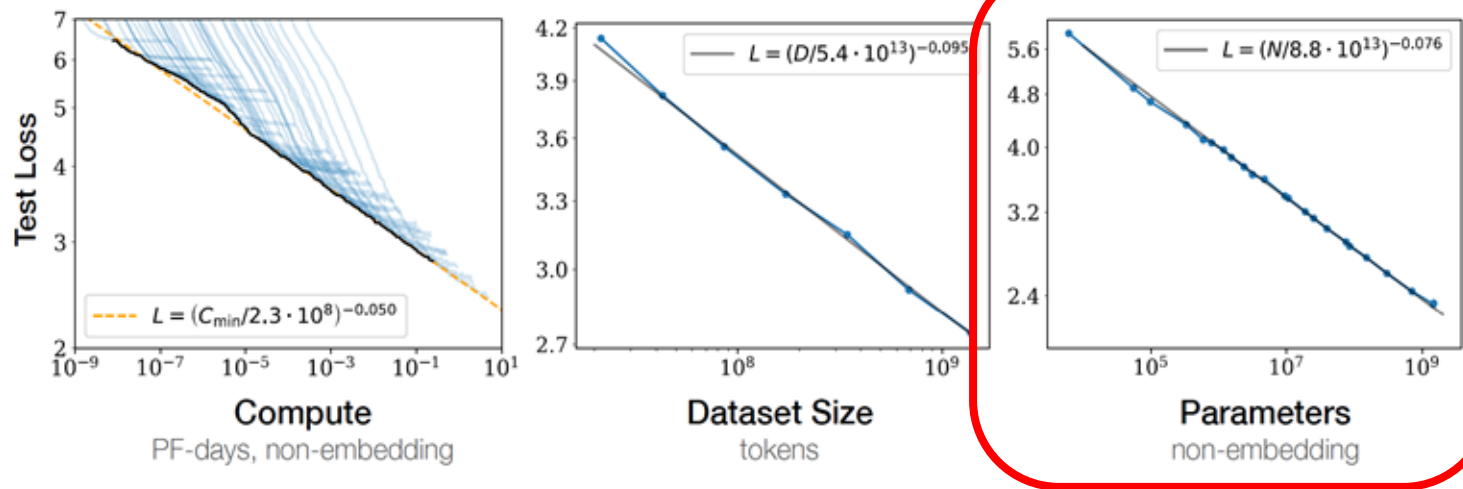
Attention Head Dimension



[Kaplan+ 2020]

- **Attention head configurations** have minimal impact on loss and can be compensated by modest additional compute

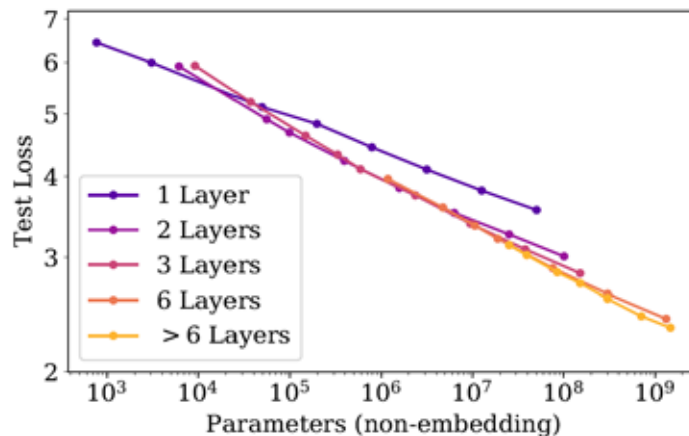
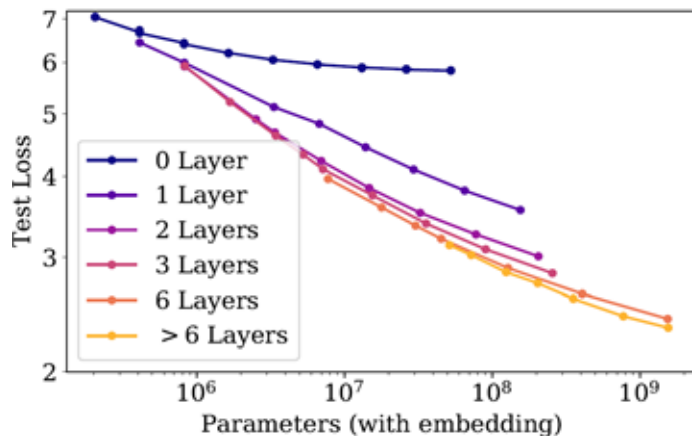
Performance with Non-Embedding Parameter Count N



- Test loss decreases as a smooth power law with **model size**, making parameter count a primary driver of performance [Kaplan+ 2020]

Which Parameter to scale?

Parameters (with embedding) vs Parameters (non-embedding)

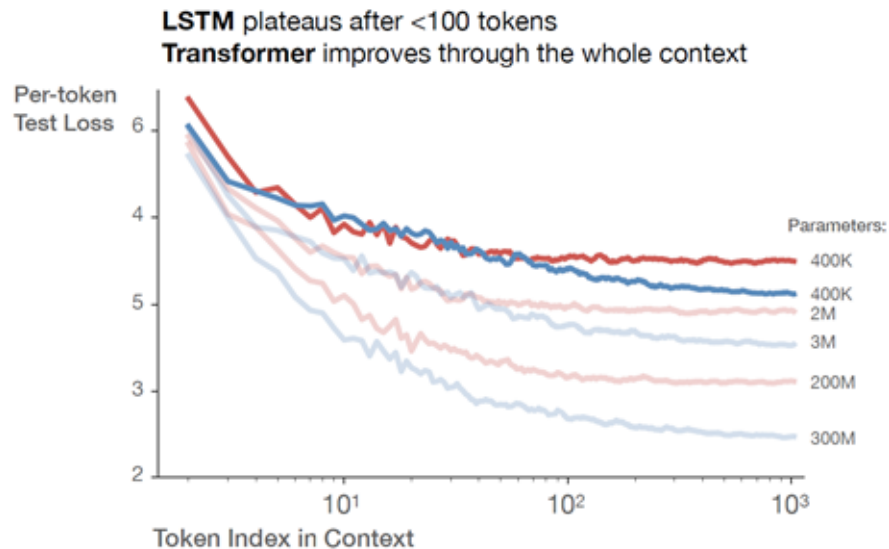
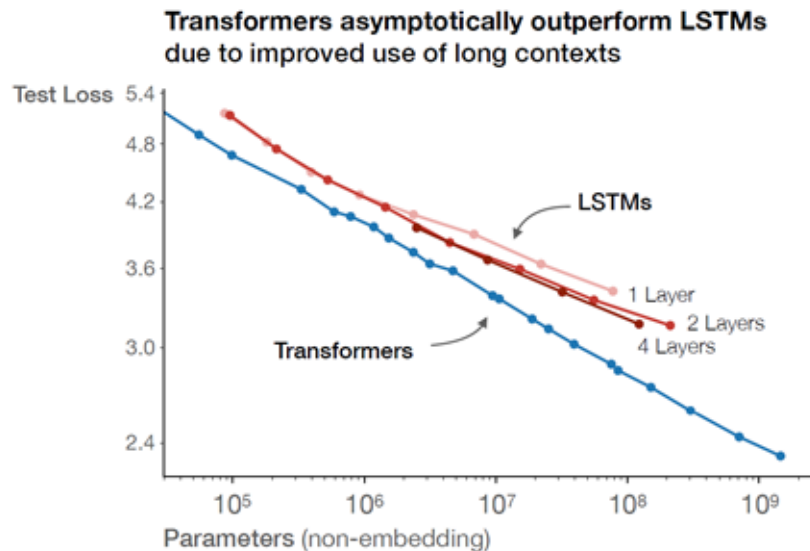


[Kaplan+ 2020]

- **Non-embedding parameter** count N is the relevant scale for performance, while **embedding size** has little impact beyond shallow models

Architecture affects Parameter Efficiency

Transformers vs LSTMs

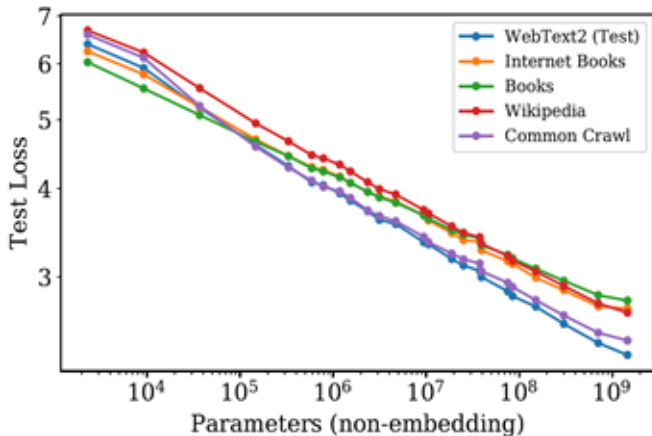


- Non-embedding parameters translate more effectively into performance gains in **Transformers** than in **LSTMs**, due to superior utilization of long-range context

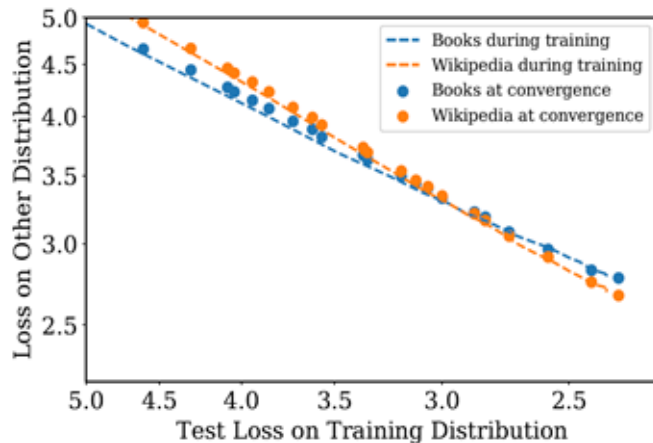
Cross-domain Performance (Transfer Learning Setting)

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Training on WebText2 → Evaluating on cross-domains



(Training is performed only on the WebText2 dataset)

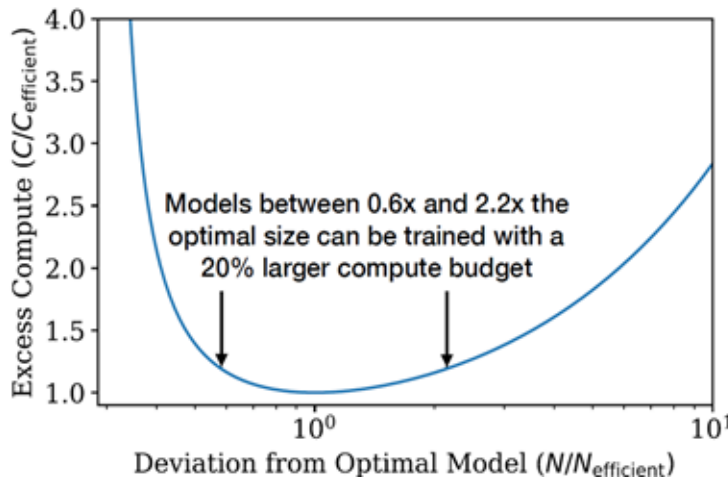


[Kaplan+ 2020]

- As model size increases, improvements on the training distribution translate almost directly into improvements on **other data distributions**

Compute-Optimal Model Behavior

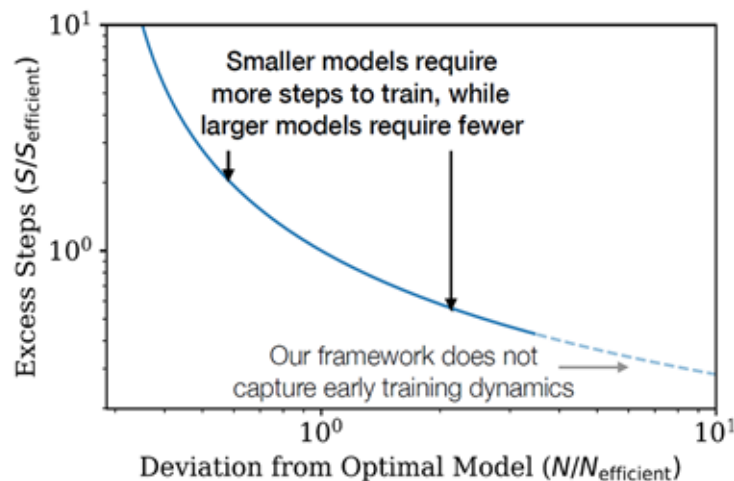
Compute-Optimal Model Size



- There exists a **compute-optimal model size** for a given C , but the optimum is broad rather than sharp [Kaplan+ 2020]

Compute-Optimal Model Behavior

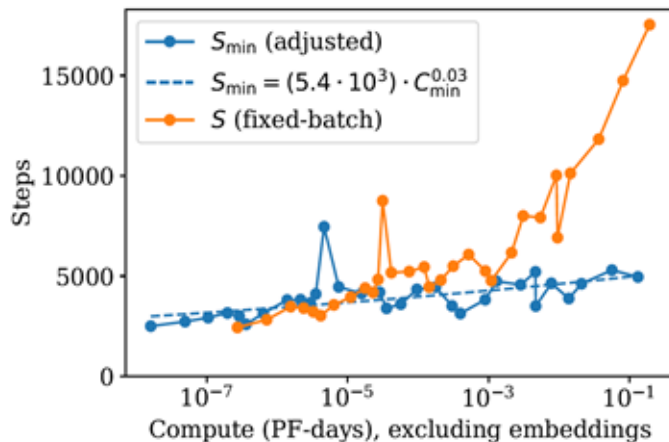
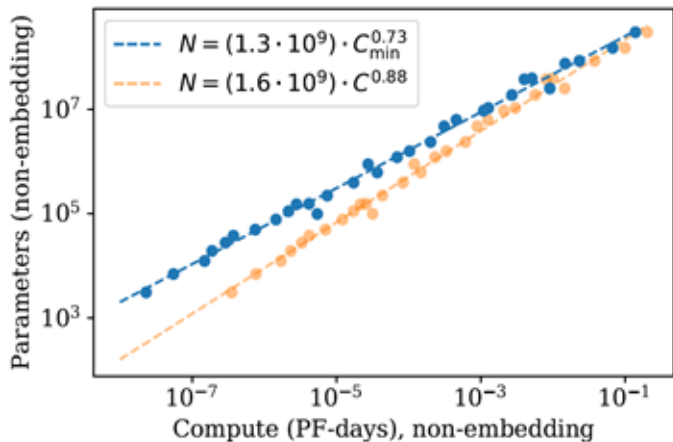
Training Dependence on Model Size



[Kaplan+ 2020]

- **Smaller models need many more steps, while larger models converge faster**

Compute-Optimal Allocation: Model Size vs. Training Steps

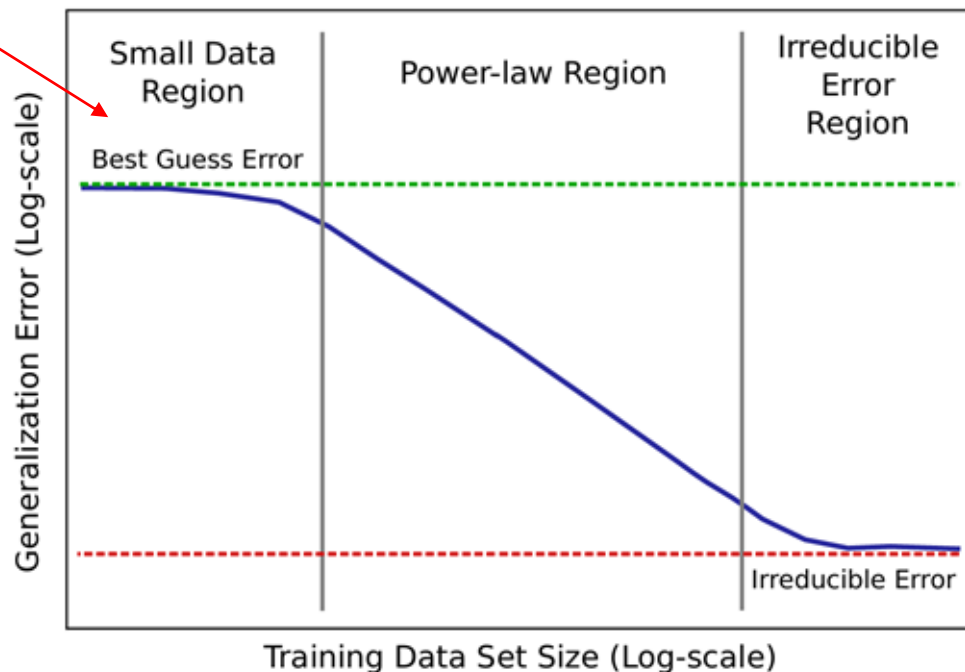


- As compute scales, it is more efficient to **scale model size** than to extend **training duration** [Kaplan+ 2020]
- Later Chinchilla argued that Kaplan-style scaling used models that were too large and insufficiently trained, and that better performance could be achieved by using smaller models trained on much more data for the same compute budget.

5. Limitation of Power-Law Scaling

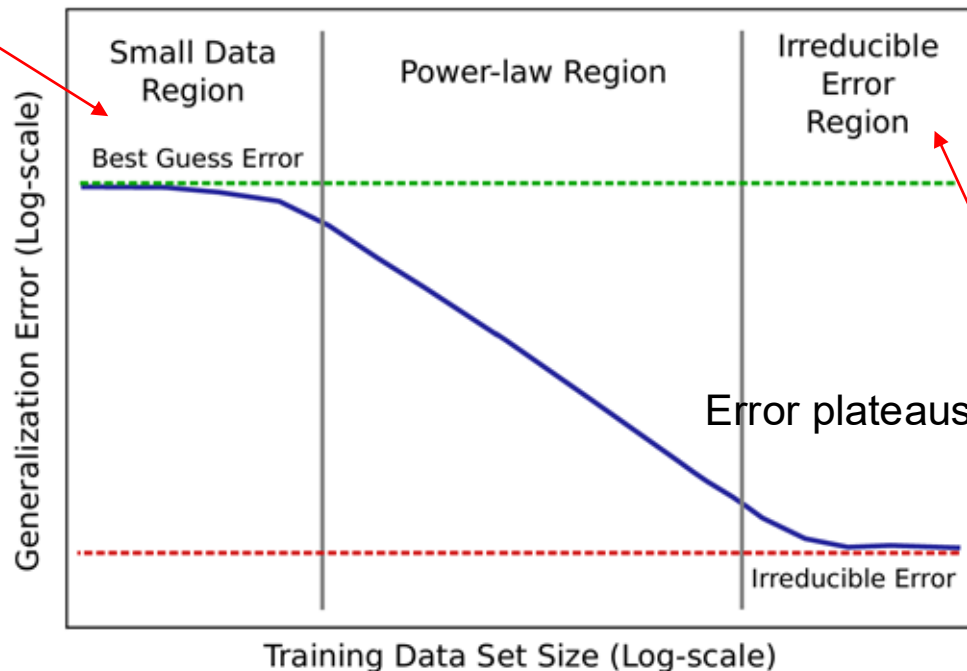
Power-law scaling is regime-dependent

Optimization and finite-size effects dominate;
scaling is unstable



Power-law scaling is regime-dependent

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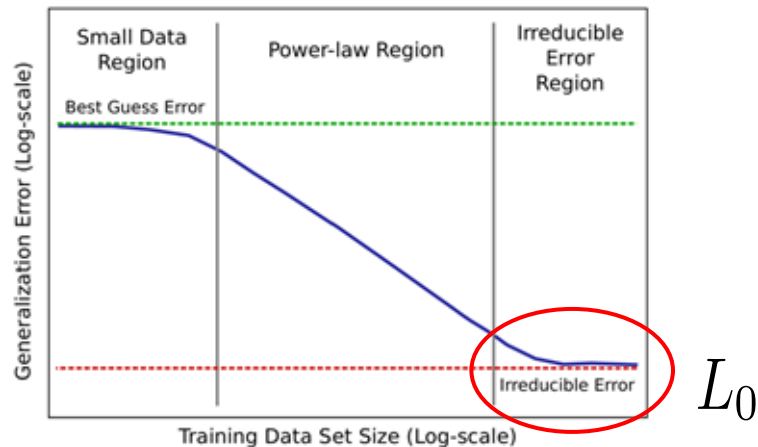
Error plateaus due to task or data limits

What is irreducible Error?

- **Irreducible Error:**

- Performance limit imposed by **intrinsic uncertainty** in the data distribution or observation process
- Ex) Bayes Error
(Theoretical minimum error achievable by any classifier)

- **Scaling Law Perspective**



[1] Vo-Van et al. "A New Strategy for Short-Term Stock Investment Using Bayesian Approach" JAFEB, 2022

[2] Hestness, Joel, et al. "Deep Learning Scaling Is Predictable, Empirically." *arXiv*, 2017.

Failure modes of power-law scaling

- **Data Exhaustion Cliff**

- Occurs when data scaling relies on **lower-quality or distribution-shifted data** after high-quality sources are exhausted
- Performance **saturates or degrades** despite increasing data volume

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- **Precision / System Cliff**

- Caused by **numerical precision limits** or **system-level constraints** (e.g., FP16/BF16)
- Scaling no longer yields gains due to **hardware or optimization bottlenecks**

Summary

- **Scaling Laws enable predictable performance extrapolation from small to large models**
- **Loss follows a power-law trend with compute, data size, and model parameters**
- **Scaling is regime-dependent and breaks due to irreducible error and practical cliffs**

Questions

- **How can we establish scaling laws cost-effectively?**
- **How reliably can small-scale experiments extrapolate to large-scale performance across different scaling regimes?**
- **Why do simple power-law relations emerge so consistently across models, data scales, and compute budgets?**
- **Where do scaling laws fundamentally break, and how can irreducible error and practical cliffs be anticipated in advance?**

What's next?

- Scaling Laws II